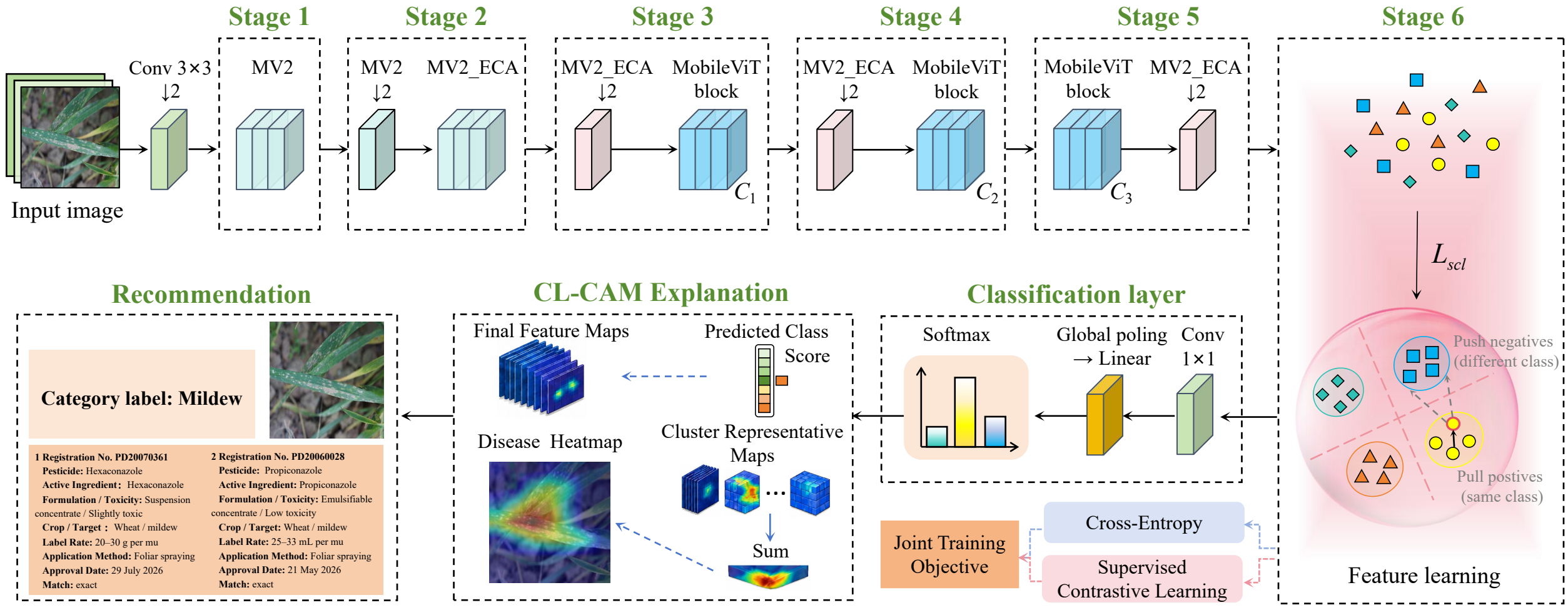

Stage 1
Stage 2
Stage 3
Stage 4
Stage 5
Stage 6
Input image
Conv 3×3 ↓2
MV2
MV2 ↓2
MV2_ECA
MV2_ECA ↓2
MobileViT block
C_1
MV2_ECA ↓2
MobileViT block
C_2
MobileViT block
MV2_ECA ↓2
C_3
L_{scl}
Push negatives (different class)
Pull postives (same class)
Feature learning
Classification layer
Softmax
Global poling → Linear
Conv 1×1
CL-CAM Explanation
Final Feature Maps
Predicted Class Score
Disease Heatmap
Cluster Representative Maps
Sum
Recommendation
Category label: Mildew
1 Registration No. PD20070361
Pesticide: Hexaconazole
Active Ingredient: Hexaconazole
Formulation / Toxicity: Suspension concentrate / Slightly toxic
Crop / Target : Wheat / mildew
Label Rate: 20–30 g per mu
Application Method: Foliar spraying
Approval Date: 29 July 2026
Match: exact
2 Registration No. PD20060028
Pesticide: Propiconazole
Active Ingredient: Propiconazole
Formulation / Toxicity: Emulsifiable concentrate / Low toxicity
Crop / Target: Wheat / mildew
Label Rate: 25–33 mL per mu
Application Method: Foliar spraying
Approval Date: 21 May 2026
Match: exact
Joint Training Objective
Cross-Entropy
Supervised Contrastive Learning